Republic of the Philippines
PARTIDO STATE UNIVERSITY
Research and Development Division
Camarines Sur

CERTIFICATION

To whom it may concern,

This is to certify that **Dr. Joni Neil B. Capucao** of Partido State University, College of Arts and Sciences has completed research approved by the University with the following details to wit:

Research Title	Proponent(s)	Original Approved Duration	Date of Completion	Authorship (Main Author, Co-Author)
Project Title: Implementation of progressive web application as platform for internal document warehousing: The effects of using web application in dynamic business process				
Study 2. Development of progressive web applications for internal document warehousing	Dr. Joni Neil B. Capucao Nicolas A. Pura Kennedy C. Cuya Jholan O. Sabaria Krismabelle R.Felizmenio Froiland B. Peñalosa	May 2021-April 2022	June 10, 2022	Main Author

This certification is issued for what ever reference/purpose it may served to the aboved requestor.

Issued this 10th day of June 2022 at Partido State University Research and Development Division.

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Director, Research and Development

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Vice President for Research, Extension and Knowledge Management

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DATE: 6/15/22

Development of progressive web applications for internal document warehousing

Partido State University
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2022

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RESEARCH STUDY TERMINAL REPORT

BASIC INFORMATION

Study Title	: Development of Progressive Web Application for Internal Document Warehousing
Researcher (s)	: Joni Neil B. Capucao (Study Leader), Nicolas A. Pura, Kennedy C. Cuya, Jholan Sabaria, Krismabelle Felizmenio, Froiland Peñalosa
Campus/College	: College of Arts and Sciences Partido State University
Approved Budget	: Php65,296.40
Duration	: May 1, 2021 to April 30, 2022

EXECUTIVE SUMMARY

The changing technological pressure on management has led to an increasing interest in improving operational processes not merely to enhance performance but as well as staying current. Government organizations in the Philippines, particularly in the academe sector, such as the State Universities and Colleges, have experienced various management difficulties in implementing efficient automated enabled technologies. In response to these needs, the development of a web-based system for internal document warehousing system will enable the SUCs to automate the existing manual processes in document management, become more transparent, and provide efficient public service.

Partido State University (PSU), a state university from the Fourth Congressional District of Camarines Sur, Philippines has the same concern in following government mandates for a quality management system. However, most of the execution of processes in the University are manually done that often led to some specific problems like difficulty in tracking documents. The absence of a web-based internal document warehouse system in the different responsibility units of the University creates this difficulty.

The development of electronic hub system for the Partido State University is but a responsive, professional commitment of the proponent in addressing this gap.

The general objective of the present study is to design and develop a web application for Internal Document warehousing to enhance the tracing of document's whereabouts for Partido State University, Goa, Camarines Sur. To address this need, the researcher answered the following questions. 1.) What system requirements (hardware, software, peopleware) can be adapted in establishing internal document warehouse; 2) What web services can be engineered that will serve as the basis for developing frameworks of the project(s); 3) What are the models that can be developed and applied to create the progressive web application.

The developed prototype internal document warehouse system was first subjected, through preliminary interviews with twenty-eight (28) office heads, to draw out the information requirements for the system's essentials that could adapt to the present demands. Specifically, they identified four (3) data requirements and three (3) information and communication technology (ICT) requirements. Along with the highest number of controlled documents regularly handled, Registrar and School of Graduate Studies have fifty-four (54) controlled documents. Along with the highest number of master lists of logbooks

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usually accomplished, the Registrar had twenty (20) logbooks. Nevertheless, the Registrar showed sixteen thousand four hundred forty-one (16,441), or 15.19 percent monitored documents. In terms of the information and communication technology requirements, there were three major components identified (system users, software, and hardware requirements).

The evaluations of information resources were now established. The researcher then designed diagrams or models for the validation of the eleven (11) member technical circle whose composition included information technology (IT) and software user experts. In terms of activity diagram, all eleven (11) or 100 percent jurors affirmed that there is one main activity needed. In terms of use case diagram, all eleven (11) or 100 percent jurors affirmed that five (3) system users (admin, stakeholders, staff/unit heads) were required. In terms of the deployment diagram, still, all eleven (11) or 100 percent jurors affirmed that use of Internet Browsers and Email were indispensable in the system features.

The developed web application for Internal Document warehousing was assessed by the end-users. As regard to usability level the following ratings were obtained: for usefulness, Mw 4.40 interpreted Very Good; for ease of use, Mw 4.37 interpreted Very Good; for ease of learning, Mw 4.44 interpreted Very Good; and for satisfaction, Mw 4.42 interpreted Very Good. Test of significant difference under usability level indicated the following p-values: 0.3806 for usefulness; 0.2627 for ease of use; 0.3818 for ease of learning; and 0.1123 for satisfaction, at .05 level of significance showed all four (4) indicators not significant.

Some information requirements were identified. These were the list of controlled documents, the master list of documents, monitored documents, code structures, peopleware, software, and hardware requirements. These data were identified as indispensable information requirements by the twenty-eight (28) office heads that it should be adopted in the design for internal document warehousing system so that the new system could address their generic needs.

There were three (3) models developed for PSU in internal document warehousing system – activity diagram, use case diagram, and deployment diagram. These features were chosen by technical experts and expert users within the local context of PSU who actually have direct involvement and hence, their contributory stance is overriding.

The respondents were satisfied with the developed internal document warehousing system. Tests of significant difference for usefulness, ease of use, ease of learning, and satisfaction level showed no significant difference and therefore respondents confirmed that the developed system met the basic requirements of the users. The internal document warehousing system, not only conformed to the generic needs of the staff's users, but also offered to the whole students, faculty, and other clientele credible and transparent transactions.

Some management concerns that are not part of document tracking and aggregating of expenses relative to information requirements should be attended later after the regular operation of the proposed prototype system.

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ELECTRONIC HUB SYSTEM FOR PARTIDO STATE UNIVERSITY

PROJECT TITLE

IMPLEMENTATION PLAN

Objective:
Establish an internal document warehousing system to centralize, store, and manage university documents efficiently.

1. **Preliminary Research and Needs Assessment (June 11 - June 20)**
   - Conduct a needs assessment with relevant stakeholders.
   - Evaluate the current document storage system.
   - Determine system requirements and functionality.
2. **Vendor Evaluation and Selection (June 21 - July 5)**
   - Identify potential vendors and software solutions.
   - Set evaluation criteria based on needs assessment.
   - Schedule demos and evaluate software functionalities.
   - Select a suitable vendor and finalize the contract.
3. **Infrastructure Setup (July 6 - July 20)**
   - Set up required servers and databases.
   - Ensure security measures (firewalls, encryption, etc.).
   - Purchase necessary licenses and software tools.
4. **Customization and Integration (July 21 - August 10)**
   - Customize the software as per university requirements.
   - Integrate the system with existing university tools (if applicable).
   - Perform data migration.
5. **User Training (August 11 - August 20)**
   - Organize training sessions for key users.
   - Provide hands-on training and software documentation.
   - Address queries and provide troubleshooting guidelines.
6. **Soft Launch (August 21 - August 31)**
   - Launch the system for a select group of users.
   - Gather feedback and make necessary tweaks.
   - Monitor system performance.
7. **Official Launch and Promotion (September 1 - September 10)**
   - Officially launch the system to all relevant users.
   - Promote usage through emails, workshops, and orientations.
   - Provide a helpline or support desk for user queries.
8. **Ongoing Maintenance and Support (September 11 onwards)**
   - Continually monitor system performance.
   - Provide regular software updates.
   - Address any issues promptly.

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GANTT CHART OF ACTIVITIES

Activity	June	July	August	September
Preliminary Research & Needs				
Vendor Evaluation & Selection				
Infrastructure Setup				
Customization & Integration				
User Training				
Soft Launch				
Official Launch and Promotion				
Ongoing Maintenance and Support				

Submitted by:

JONI NEIL B. CAPUCAO, DIT
Faculty Researcher

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FACULTY POSITION RECLASSIFICATION FOR SUCs
(DBM-CHED Joint Circular No. 3, series of 2022)

CERTIFICATION OF PERCENTAGE CONTRIBUTION

(Research Output Translated into Project/Policy or Product with Multiple Authors)

Title of Research:
Title of Project/Policy/Product: **Development of Progressive Web Applications for Internal Document Warehousing Electronic Hub System for Partido State University**

Instruction: Supply ALL the names of the authors involved in the research output translated inyp project/policy or product and indicate the contribution of each author in percentage. Each author shall sign the conforme column if he/she agrees with the distibution. The conforme should be signed by all the authors in order to be considered. Please prepare a separate Certification for each output.

	Name	Current Affiliation	% Contribution	Conforme *(Sign if you agree with the % distribution)*
1	Dr. Joni Neil B. Capucao	Partido State University	50%	
2	Kennedy C. Cuya	Partido State University	30%	
3	Nicolas A. Pura	Partido State University	5%	
4	Jholan O. Sabaria	Partido State University	5%	
5	Krismabelle R. Felimenio	Partido State University	5%	
6	Froiland B. Peñalosa	Partido State University	5%	
7				
	**should have a total of 100%*		100%	

Prepared by:

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(If the one certifying is the co-author, the next higher academic official shall sign the certification)

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